2017 Ph H2 Q9

Section: Particles and Waves

Topic: Nuclear Reactions

A nuclear fusion reaction is represented by the equation:

$$^{2}_{1}\mathrm{H} + ^{3}_{1}\mathrm{H} \rightarrow ^{4}_{2}\mathrm{He} + ^{1}_{0}\mathrm{n}$$

(a)

Show that the total charge and total mass number are conserved in this reaction.

✅ Check **mass numbers** (top numbers):

$$2 + 3 = 5 \quad \text{on left}, \quad 4 + 1 = 5 \quad \text{on right}$$

✅ Check **charges** (bottom numbers):

$$1 + 1 = 2 \quad \text{on left}, \quad 2 + 0 = 2 \quad \text{on right}$$

✔️ Both mass number and charge are conserved.

> ⚠️ *Revision tip:* Always check nuclear equations for **conservation of charge and nucleon number** (protons + neutrons).

(b)

The mass of a deuterium nucleus is 3.343×10^{-27} kg.
The mass of a tritium nucleus is 5.006×10^{-27} kg.
The mass of the products is 8.374×10^{-27} kg.
Calculate the energy released in the reaction.

Step 1: Find **mass difference (Δm)**

$$\Delta m = (3.343 + 5.006) - 8.374 = 8.349 - 8.374 = -0.025 \times 10^{-27}\ \text{kg}$$

Take the magnitude:
$\Delta m = 2.5 \times 10^{-29}\ \text{kg}$

Step 2: Use $E = mc^2$

$$E = 2.5 \times 10^{-29} \times (3.0 \times 10^{8})^2 = 2.5 \times 9 \times 10^{-13} = \ 2.3 \times 10^{-12}\ \text{J}$$

✅ *Revision tip:*

- Be careful with mass units — always in **kg**
- Fusion releases energy because **mass is lost and converted to energy**
